

Exercise Set 7

27 October 2025

1 DPLL with theories

1.1 DPLL(T)

- Use DPLL(T) to solve the following

1. $a + 1 \leq b \vee a \leq 1$
2. $a + 1 > b \vee a > b + c$
3. $b > 0 \vee a + b > b + b$
4. $a \leq b + c \vee b > 0 \vee c > 0$

Which theory do you use? Is the system satisfiable?

- Again use DPLL(T) to solve the following

1. $a = b \vee f(b) = f(d)$
2. $a \neq b \vee f(c) = f(b)$
3. $a \neq b \vee g(f(b)) \neq g(f(a)) \vee g(f(a)) \neq g(f(c))$
4. $f(b) \neq f(d) \vee g(f(a)) = g(f(c))$
5. $f(b) \neq f(d) \vee f(c) = f(b)$
6. $a = c \vee a = d$
7. $a = b \vee f(a) \neq f(d)$

Which theory do you use? Is the system satisfiable?

1.2 Theory for arrays

Let us introduce the theory on the arrays. We use the two functions

- `.store(a,i,e)`: returns the same array as `a`, with the element `e` placed at index `i`
- `.select(a,i)`: returns the element of `a` at index `i`

Express the following statements as FOL formulas

- If an element e is stored at index i in an array a , then what it is selected afterwards at index i is actually e
- If an element e is stored at index i in an array a , then what it is selected afterwards at an index different from i is actually what was in a before the **store** operation
- If two arrays contain the same elements at the same indices then they are the same

1.3 Little test code

You are given the following two pieces of code. a is an array, i and j are two valid indices and b and c are two integer values

```
array1(a,i,j,b,c) =
  a[i] := b;
  a[j] := c;
```

```
array2(a,i,j,b,c) =
  a[j] := c;
  a[i] := b;
```

Are they equivalent?

- Express **array1** with the **theory for arrays** of previous exercise (same for **array2**)
- Intuitively, for which conditions we have that the result of **array1** is equal to the result of **array2**?
- Once you found the condition, you can now try to code this in **why3**, adding it and try to obtain the green V.